Yoolkyu Park

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Academic Interests

Biostatistics; Computational Biology; Medical Imaging; Spatial Statistics; Topological Data Analysis

Education

The University of North Carolina at Chapel Hill

B.S., Mathematics, Statistics and Analytics (Double Major) Aug 2025 – May 2027
GPA: 3.57/4.00

The University of Texas at Dallas

Coursework in Mathematics and Data Science (Transferred) Aug 2023 – May 2025
GPA: 3.84/4.00

Publications and Preprints

1. **Y. Park**, F. Wu, X. Feng, S. Yang, E. H. Wang, B. Yao, C. Moon, G. Xiao, and Q. Li(2025). Spatial Analysis for AI-segmented Histopathology Images: Methods and Implementation. Submitted for review; preprint available. (arXiv:2512.06116)

Research Experience

CompCy Lab, Department of Computer Science, UNC Chapel Hill

Undergraduate Researcher Feb 2026 – Present
Advisor: Dr. Natalie Stanley

- Investigating multiplex imaging mass cytometry (IMC) data to support spatial analyses aimed at characterizing neuroimmune response patterns.

Qiwei Lab, Department of Mathematical Sciences, UT Dallas

Undergraduate Researcher Aug 2024 – Oct 2025
Advisor: Dr. Qiwei Li

- Collaborated with an external development team to implement the analytical backend for a web-based spatial analysis tool derived from SASHIMI.
- Translated research specifications into deployable analytical modules; maintained version-controlled code and documentation using Git.

Outreach

- Poster presentation: "Spatial Analysis for AI-segmented Histopathology Images: Methods and Implementation," EAC–ISBA 2025, Daegu, South Korea. *Best Student Poster Award*.

[1] Updated March 4, 2026

- Poster presentation: "Spatial Analysis for AI-segmented Histopathology Images: Methods and Implementation," Undergraduate Research Symposium, 2025, UT Dallas, Richardson, TX.

Awards

- Dean's List – The University of North Carolina at Chapel Hill, 2026.
- Best Student Poster Award – East Asia Chapter, International Society for Bayesian Analysis Conference (EAC–ISBA), 2025.
- Travel Award (selected) – Statistics Undergraduate Research Experience, SRCOS 2025.
- Full-time Undergraduate Research Studentship ($3,000) – The University of Texas at Dallas, 2024.
- Dr. John Jagger Academic Excellence Scholarship – The University of Texas at Dallas, 2024.
- Dean's List – The University of Texas at Dallas, 2024.

Selected Projects

SASHIMI: Spatial Analysis for Segmented Histopathology Images using Machine Intelligence

- Developed an end-to-end pipeline for large-scale cell-level spatial data (37.5M+ observations), transforming segmented histopathology images into marked point pattern representations and structured spatial feature tables for downstream inference.
- Applied functional principal component analysis (FPCA) to high-dimensional spatial descriptors to obtain low-dimensional representations of cell–cell interaction patterns while preserving interpretability linked to tissue architecture.
- Modeled associations between spatial features and patient survival outcomes using Cox proportional hazards models with cross-validation to assess stability and generalizability of inferred risk patterns.
- Wrote the first-author manuscript and supplementary materials, documenting methodological assumptions and workflows to support transparency and reproducibility.

Technical Skills

- **Methods:** Spatial point process analysis; survival analysis (Cox proportional hazards); functional data analysis / FPCA; high-dimensional feature engineering; cross-validation and model evaluation.
- **Programming:** Python, R, SQL (scikit-learn, pandas, numpy, matplotlib).
- **Software Practices:** Git/GitHub; reproducible workflows.
- **Writing:** LaTeX; Markdown; technical documentation.